Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Describe how a fixed loop is used in the program.

- for b0 = 1 to 20 … next b0 surrounds the sequence
- After 20 repeats the program moves on to end

3 Quick check

- a) 2 s
- b) Diamond
- c) It can be reprogrammed without changing the circuit
- d) Fewer components, so more reliable

4 Key terms

- 1 – A
- 2 – E
- 3 – C
- 4 – B
- 5 – D

5 Draw the flowchart symbol for a decision and for an output.

- Decision: diamond with yes/no branches
- Output: parallelogram (or rectangle with "high/low")

6 Explain what is meant by ‘pause 500 (PBASIC)’.

- A 0.5 s delay: pause is in milliseconds.

7 Describe the advantage of a microcontroller.

- Reprogrammed without changing the circuit; fewer components.